ANSWER KEY					AT#21 (UNITS & DIMENSIONS, ERRORS)				
Q.	1	2	3	4	5	6	7	8	9
A.	4	3	2	3	2	4	3	3	4
Q.	10	11	12	13	14	15	16	17	18
A.	3	2	2	1	3	4	3	3	1
Q.	19	20	21	22	23	24	25	26	27
A.	1	1	4	4	2	1	3	4	3
Q.	28	29	30	31	32	33	34	35	36
A.	2	1	4	3	2	1	4	4	4
Q.	37	38	39	40	41	42	43	44	45
A.	2	1	4	3	3	2	4	4	3

Hints And Solution

1. ANS – 4

$Q = I^2Rt$,

Required error is $2 \times 2\% + 1\% + 1\% = 6\%$

2. ANS – 3

$$\omega = \frac{1}{\sqrt{LC}} = \frac{2\pi}{T}$$

3. ANS – 2

20 VSD= 16 MSD= 16mm

$\Longrightarrow$ 1 VSD $= \frac{4}{5}$ mm

Least count =1 MSD -1 VSD = 1 - $\frac{4}{5}$ = 0.2 mm

4. ANS – 3

$$P = \frac{a - t^2}{bx} = \frac{a}{bx} - \frac{t^2}{bx}$$

By the principle of homogeneity

$$[P] = \frac{force}{area} = \frac{a}{b[x]}$$

$ML^{-1}T^{-2} = \frac{a}{b}\left[\frac{1}{L}\right]$

Or $\frac{a}{b} = MT^{-2}$

5. ANS – 2

Random error is inversely proportional to the number of observations

So, $Error\ \alpha \frac{1}{observations}$

$$\frac{E_1}{E_2} = \frac{O_2}{O_1}$$

$$E_2 = \frac{100\,x}{400} = \frac{x}{4}$$

6. ANS – 4

The dimension of A is not equal to the dimension of C.

Hence it is not possible to subtract AC from A^2

7. ANS – 3

$$coefficient\ of\ friction = \frac{\text{Applied force}}{\text{Normal reaction}}$$

$$= \frac{[MLT^{-2}]}{[MLT^{-2}]} = \text{no dimensions}$$

$$\text{Unit} = \frac{N}{N} = \text{no unit}$$

8. ANS – 3

Mean time period T = 2.00 s and Mean absolute error = ΔT = 0.05 s

To express maximum estimate of error, the time period should be written as
$(2.00 \pm 0.05)\ s$

9. ANS – 4

$$V = \frac{\pi d^2 h}{4} = \frac{\pi}{4}(12.6)^2(34.2)cm^3 = 4260cm^3$$

Hence, maximum error in volume is given by

$$\frac{\Delta V}{V} = \frac{2\Delta d}{d} + \frac{\Delta h}{h}$$

$$\frac{\Delta V}{4260} = 2\left(\frac{0.1}{12.6}\right) + \left(\frac{0.1}{34.2}\right)$$

$$\Delta V = 80\ cm^3$$

$$V = 4260 \pm 80\ cm^3$$

10. ANS – 3

Pitch =0.5 mm, L.C = $\frac{0.5}{50}$ = 0.01 mm
M.S.R. = 2

Neck reading = 25

d = $2 \times 0.5 + 25 \times 0.01$ = 1.25 mm

Considering zero error

Here zero error is = $+5 \times 0.01 = 0.05$ mm

So actual diameter = $d - 0.05 = 1.20$ mm

11. ANS – 2

$$KE = \frac{1}{2}\frac{p^2}{m}$$

$$KE_1 = \frac{1}{2}\frac{p^2}{m}$$

$$p_1 = \frac{6}{5}p$$

$$KE_2 = \frac{1}{2}\left(\frac{36}{25}\right)\frac{p^2}{m}$$

$$\Delta KE = KE_2 - KE_1$$

$$= \frac{1}{2}\left[\frac{\left(\left(\frac{6}{5}p\right)^2 - p^2\right)}{m}\right] = \frac{11}{25}\frac{p^2}{m}$$

Percentage change in kinetic energy

$$= \frac{\Delta KE}{KE_1} \times 100 = \frac{11}{25} \times 100 = 44\%$$

12. ANS – 2

According to the principle of dimensional homogeneity

$[p] = [\alpha / V^2]$

$[\alpha] = [ML^{-1}T^{-2}][L^6] = [ML^5T^{-2}]$

Hence the unit of $\alpha = gm \times cm^5 \times s^{-2} = dyne \times cm^4$

13. ANS – 1

Force F = $[M^1L^1T^{-2}] = 100N$.......(i)

Length L = $[L] = 10$ m ....(ii)

Time t = $[T] = 100$ s ....(iii)

Substituting values of L and T from equations (ii) and (iii) in equation (i), we get

$$M \times 10 \times (100)^{-2} = 100$$

$$\frac{M \times 10}{100 \times 100} = 100 \Rightarrow M = 10^5$$

Unit of mass in new system = 10^5kg

14. ANS – 3

$$Percentage\ error = \frac{least\ count}{Total\ time} \times 100$$

$$\frac{0.2}{25} \times 100 = 0.8\%$$

15. ANS – 4

$$\frac{B^2}{2\mu_0} = Energy\ density = \frac{Energy}{Volume}$$

$$\Rightarrow \frac{[ML^2T^{-2}]}{[L^3]} = [ML^{-1}T^{-2}]$$

16. ANS – 3

Let m α $C^xG^yh^z$

By substituting the following dimensions:

$[C] = LT^{-1}; [G] = [M^{-1}L^3T^{-2}]$

and$[h] = [ML^2T^{-1}]$

Now, comparing both sides, we get

X= ½ , y=-1/2, z=+1/2

So, m α $c^{1/2}G^{-1/2}h^{1/2}$

17. ANS – 3

P= F/A=$F/\pi r^2$

$$\frac{\Delta P}{P} \times 100 = \frac{\Delta F}{F} \times 100 + \frac{2\Delta r}{r} \times 100$$

$$= 5 + 2(3) = 5 + 6 = 11\%$$

18. ANS – 1

$$\frac{\Delta s}{s} = 2 \times \frac{\Delta r}{r} \text{ and } \frac{\Delta V}{V} = 3 \times \frac{\Delta r}{r}$$

$$\therefore \frac{\Delta v}{v} = \frac{3}{2}\frac{\Delta s}{s}$$

$$\frac{\Delta v}{v} = \frac{3}{2}\alpha$$

19. ANS – 1

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} \ or\ c^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$The\ dimensional\ formula\ is\ , \left[\frac{1}{\mu_0 \epsilon_0}\right]$$

$$= [M^0L^2T^{-2}]$$

20. ANS – 1

$$Energy\ \alpha\ [Force]^x[Acceleration]^y[Time]^z$$

$$[ML^2T^{-2}] = [ML\,T^{-2}]^x\,[L\,T^{-2}]^y[T]^z$$

$$On\ solving\ we\ get, x = 1, y = 1, z = 2$$

$$[Energy] = [F^1A^1T^2]$$

21. ANS – 4

$$R = \frac{V}{I}$$

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I}$$

$$\therefore \%eror\ in\ R = \frac{2}{50} \times 100 + \frac{0.2}{5} \times 100 = 8\%$$

22. ANS – 4

Least count of screw gauge = 5µm

$$\text{Least count} = \frac{Pitch}{No\ of\ division\ on\ circular\ scale}$$

$$N = \frac{1mm}{5\mu m}$$

$$N = 200$$

23. ANS – 2

Frequency $[M^0L^0T^{-1}] = [M]^X[ML^0T^{-2}]^Y$

$= [M]^{X+Y}\,[L]^0[T]^{-2Y}$

Comparing the power on M, L, and T

$$-2Y = -1 \quad Y = \frac{1}{2}$$

$$And\ x + y = 0\ \therefore x = -y = -\frac{1}{2}$$

24. ANS – 1

$$h = G^X L^y E^z$$

$$[M^1L^2T^{-1}] = [M^{-1}L^3T^{-2}]^x[M^1L^2T^{-1}]^y[M^1L^2T^{-2}]^z$$

Comparing the power we get

1 = -x + y + z … (i)

2 = 3x + 2y + 2z … (ii)

-1 = -2x –y – 2z … (iii)

On solving eq (i), (ii) & (iii),

We get, x = 0

25. ANS – 3

Pitch = $\frac{1}{2}$ mm

Least count = $\frac{0.5}{50} = 0.01$ mm

Reading = Main scale reading + Circular scale reading + Magnitude of negative error.

$= \left[2 \times \frac{1}{2} + 20 \times 0.01 + (50 - 30) \times 0.01\right)$

= 1.4 mm

26. ANS – 4

$$x = R\frac{l}{100 - l}$$

$$\frac{\Delta x}{x} = \frac{\Delta l}{l} + \frac{\Delta l}{100 - l}$$

$$\frac{\Delta x}{x} \times 100 = \left(\frac{1}{50} + \frac{1}{50}\right) \times 100 = 4\%$$

27. ANS – 3

Viscous force, F = 6πηrv

$$\eta = \frac{F}{6\pi r v} \Rightarrow [\eta] = \frac{[F]}{[r][v]} = \frac{[MLT^{-2}]}{[L][LT^{-1}]}$$

$= [ML^{-1}T^{-1}]$

28. ANS – 2

29. ANS – 1

According to the question

Muscle x Speed =Power

Muscle= $\frac{\text{power}}{\text{speed}}$ = MLT^{-2}

30. ANS – 4

$$t = \frac{D_1 - D_2}{2} = \frac{4.23 - 3.87}{2} = 0.18\ cm$$

$$\Delta t = \left(\frac{0.01 + 0.01}{2}\right) = 0.01 cm$$

$\Rightarrow t = (0.18 \pm 0.01)$ cm

31. ANS – 3

32. ANS – 2

$$\frac{\Delta g}{g} \times 100 = \frac{\Delta l}{l} \times 100 + 2\left(\frac{\Delta T}{T} \times 100\right)$$

$$\Delta l = 0.01\text{cm}$$

$$\Delta T = 0.02\text{s}$$

$$l = 10\text{cm}$$

$$T = 0.5\text{s}$$

$$\frac{\Delta l}{l} \times 100 = \frac{0.01}{10} \times 100 = 0.1$$

$$\frac{\Delta T}{T} \times 100 = \frac{0.02}{0.5} \times 100 = 4$$

$$\frac{\Delta g}{g} \times 100 = 0.1 + 2 \times 4 \approx 8\%$$

33. ANS – 1

$$A = l^2$$

$$l = \sqrt{A} = 10\text{m}$$

$$\frac{\Delta A}{A} = 2\frac{\Delta l}{l}$$

$$\frac{\Delta l}{l} = \frac{1}{2}\frac{\Delta A}{A}$$

$$\Delta l = \frac{1}{2}\left(\frac{\Delta A}{A}\right)l$$

$$\Delta l = \frac{1}{2}\left(\frac{0.2}{100}\right) \times 10 = 0.01\text{m}$$

34. ANS – 4

$[E] = [ML^2T^{-2}]$

$[J] = [ML^2T^{-1}]$

$[G] = [M^{-1}L^3T^{-2}]$

$$\frac{[E][J]^2}{[M]^5[G]^2} = \frac{[ML^2T^{-2}][ML^2T^{-1}]^2}{[M^5][M^{-1}L^3T^2]^2} = [M^0L^0T^0]$$

35. ANS – 4

Dipole moment, $[P] = q.l = [I\ T\ L]$

Electric flux, $[\emptyset] = E.ds = [M\ L^3I^{-1}T^{-3}]$

Electric field = [E] = $\frac{F}{q}$ $= [ML\ I^{-1}T^{-3}]$

36. ANS – 4

CV^2 = Energy

We know that dimensional formula of energy is $[ML^2T^{-2}A^0]$

37. ANS – 2

$$F = \frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r^2}$$

$$\Longrightarrow \varepsilon_0 = \frac{q_1q_2}{4\pi Fr^2}$$

$$= \frac{[TA]^2}{[L]^2[MLT^{-2}]}$$

$$= [M^{-1}L^{-3}A^2T^4]$$

38. ANS – 1

M = Pole strength x length

= amp .metre . metre = amp metre^2

39. ANS – 4

$$N_1U_1 = N_2U_2$$

$$N_1[M_1L_1^{-3}] = N_2[M_2L_2^{-3}]$$

$$\therefore N_2 = N_1\left[\frac{M_1}{M_2}\right] \times \left[\frac{L_1}{L_2}\right]^{-3}$$

$$= 0.625\left[\frac{1g}{1kg}\right] \times \left[\frac{1cm}{1m}\right]^{-3}$$

$$= 0.625 \times 10^{-3} \times 10^6 = 625$$

40. ANS – 3

$$m = linear\ density$$

$$= mass\ per\ unit\ length = \left[\frac{M}{L}\right]$$

$$A = force == [\mathrm{MLT^{-2}}]$$

$$[B] = \frac{[A]}{[m]} = \frac{[\mathrm{MLT^{-2}}]}{[\mathrm{ML^{-1}}]} = [\mathrm{L^2T^{-2}}]$$

This is same dimension as that of latent heat.

41. ANS – 3

The frequency of Sonometer wire is

$$\nu = \frac{1}{2l}\sqrt{\frac{T}{m}}$$

Hence, maximum error in frequency is given by-

$$\frac{\Delta\nu}{\nu} = \frac{\Delta l}{l} + \frac{1}{2}\frac{\Delta T}{T} + \frac{1}{2}\frac{\Delta m}{m}$$

$$\frac{\Delta\nu}{\nu} = \frac{\Delta l}{l} = 1\%$$

Hence, frequency will decrease by 1 %

42. ANS – 2

Dimension of at = dimension of F

$$[\mathrm{at}] = [F]$$

$$[\mathrm{a}] = \left[\frac{\mathrm{F}}{\mathrm{t}}\right]$$

$$[\mathrm{a}] = \left[\frac{MLT^{-2}}{T}\right]$$

$$[\mathrm{a}] = [MLT^{-3}]$$

Dimension of bt^2 = dimension of F

$$[\mathrm{bt}^2] = [F]$$

$$[b] = \left[\frac{\mathrm{F}}{\mathrm{t}^2}\right]$$

$$[b] = \left[\frac{MLT^{-2}}{T^2}\right]$$

$$[b] = [MLT^{-4}]$$

43. ANS – 4

$$[Energy] = [ML^2T^{-2}]$$

$$[Thrust] = [ML^1T^{-2}]$$

$$[Momentum] = [ML^1T^{-1}]$$

$$[Angular\ velocity] = [M^0L^0T^{-1}]$$

44. ANS – 4

$$\text{Least count} = \frac{Pitch}{No\ of\ division\ on\ circular\ scale}$$

$$N = \frac{1}{100} = 0.01\ mm$$

$$Diametre = 0 + 52 \times 0.01 = 0.52\ mm$$

$$= 0.052cm$$

45. ANS – 3